



DIGITAL TECHNOLOGIES

ZUKUNFT. DIGITAL. NACHHALTIG.

SoSe 2026

Vergabepilot.AI:

Agentic AI for Automated Public Project Scraping

In Cooperation with



Machine Learning and COgnitive SoftwaRE (CORE) Research Group
European AI Team Projects (ATP)

Ein gemeinsames Studienprogramm der



Project in Cooperation with a fast-moving startup!

Ciconia Systems GmbH

We develop custom software solutions and AI applications for medium-sized companies. Our focus is on the rapid implementation of digital products using modern technologies—from web apps to generative AI.



Focus
AI-
Applications



Location
Stuttgart &
Cluj



Founded in
2022



Team
20+



1.Hackathon (1)

Phase 1: Manual Scraping (5h)

- Learn manual scraping techniques
- XML Path Language (XPath)
- Browser automation (e.g., *Playwright*)
- Two teams, 7,500 URLs per team
- Implement scrapers manually (**no LLMs**)

Goal: Maximize number of scraped websites

Phase 1
Manual Scraping



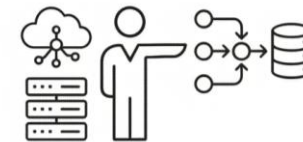
XPath, browser automation

Phase 2: LLM-Assisted Scraping

- Use different LLMs per team
- Generate scraper code with LLMs
- Test, debug, and iterate
- Keep track of iterations necessary
- How good are LLMs at generating?

Goal: Initial evaluation of LLM performance

Phase 2
Advanced & Automated Scraping



Headless browser clusters, structured data output, database loading, error handling

1.Hackathon (2)

	id	created	updated	version	url	domain	trigger_id	procedure_id
0	6a05934a-d751-4107-a1dd-dbb24ed438f7	2026-03-22 07:17:14.373 +0200	2026-03-22 07:17:25.867 +0200	1	https://www.evergabe-online.de/tenderdetails.h...	www.evergabe-online.de	b32d2a72-368a-45fd-a1f3-ef5b7aa4c101	02e5e8c9-2b40-40d0-8bd1-8b7d55f686aa
1	1342d4ba-c967-473a-8418-1f9091a813e0	2026-03-22 07:17:04.622 +0200	2026-03-22 07:17:04.983 +0200	1	https://www.dtv.de/Satellite/notice/CXP4Y89M9...	www.dtv.de	39e68c6b-1a43-487a-85af-41de8e61ef3e	d5f9a789-3e3a-4daf-b98f-469d248586dd
2	5bd3fef0-b34c-4396-94bf-b63ff81039bd	2026-03-22 07:16:31.850 +0200	2026-03-22 07:16:55.100 +0200	1	https://www.sachsen-vergabe.de/NetServer/Publi...	www.sachsen-vergabe.de	5482a1d3-1edb-42af-aff3-0a1a11970c75	a33147d0-24d4-4b23-a19f-ddad82695c16
3	e4e63f2b-6679-488e-8349-0cdbc980dae1	2026-03-22 07:16:10.828 +0200	2026-03-22 07:16:24.689 +0200	1	https://www.evergabe.de/auftraege/suche-ueber-...	www.evergabe.de	d107d276-156c-4227-add6-9347900d62a0	47f9c70e-9eba-42df-a432-19fb24b6b1c6
4	504d3c99-f8e8-4f35-8de0-3d8afce25f7a	2026-03-22 07:15:47.255 +0200	2026-03-22 07:15:59.975 +0200	1	https://vergabeportal-bw.de/Satellite/notice/C...	vergabeportal-bw.de	cfbae5e1-472a-4154-a475-dade4f3c76a3	ed5c05af-a15f-42b9-bb48-5237aca5498d

Scraper Dataset

- One scraper can be used across the same platform / websites
- Does your scraper work well on other examples of the same website?
- Older publications might not be available anymore (invalid)

OpenRouter API keys are available (50\$ starting budget per team)

2. Final Week Goals

Objective

- Explore computer-use agents (CUA)
- First experiments with CUAs
- Build understanding of agent-based interaction with tender websites

Core Tasks

- Research available agents
- Select different agents per team
- Set up browser / GUI environments
- Run experiments (document downloading)
- Analyze behavior and limitations

Goals

- Initial agent implementations
- Hands-on experience with GUI agents
- Comparative insights:
 - Manual approach (doc. downloading)
 - CUA-based approach (doc. downloading)
(Requires a manually annotated test dataset)

